Detection of Backdoor Attacks using HPCs

Method report — Problem Statement 1

1. Problem framing

Only clean-inference HPC traces are available at training time (800 rows x 3 counters: cache-references, cycles, LLC-loads). No backdoor traces are provided, so this is a strict one-class / novelty-detection problem: the model must learn the geometry of benign behaviour and flag anything sufficiently far from it, without ever having seen a positive example.

2. Data characteristics

Exploratory analysis of the 800 benign traces showed: (a) moderate-to-strong correlation between cache-references and LLC-loads (r=0.92), weaker correlation with cycles (r≈0.53); (b) cache-references and LLC-loads are close to symmetric (skew -0.67 / -0.54) while cycles is right-skewed with high kurtosis (skew 0.90, kurtosis 6.76) driven by a handful of heavy-tail rows. This motivated using a **robust** covariance estimator rather than a plain Gaussian/Mahalanobis fit, and an ensemble rather than a single parametric model.

3. Feature set: raw counters only

We evaluated two feature sets: the 3 raw counters alone, and raw + 3 derived scale-invariant ratios (cycles/cache-references, LLC-loads/cache-references, LLC-loads/cycles), motivated by the hypothesis that a backdoor trigger would leave a signature in the *relative* access pattern more reliably than in absolute magnitude. Section 9's ablation testing did not support that hypothesis once Local Outlier Factor was added to the ensemble (Section 4) — raw-only features detected a weak synthetic signal more than 2x as often. The shipped model therefore uses the 3 raw counters only; --feature-set raw_ratio remains available in the code for comparison.

4. Model: rank-normalized ensemble of four one-class detectors

Four complementary one-class models are fit on the standardized 3-d raw feature space:

- **Robust Mahalanobis distance** (MinCovDet, support_fraction=0.9) — models benign traces as a roughly elliptical cluster; the robust covariance estimate is not skewed by the outlier-heavy tail in 'cycles'.
- **Isolation Forest** (300 trees) — non-parametric, captures non-elliptical structure and is cheap at inference time.
- **One-Class SVM** (RBF kernel) — a smooth non-linear boundary around the benign manifold.
- **Local Outlier Factor** (novelty mode) — density-based and LOCAL rather than global: flags a point as anomalous if it is sparse relative to its own neighbourhood, catching structure the three global models above can miss if benign traces form more than one loose cluster (e.g. different workload phases).

Each model's raw score is rank-normalized to [0,1] against its own training-score distribution (percentile rank), then the four normalized scores are averaged into a single ensemble anomaly score. Averaging an ensemble rather than trusting one model is a deliberate hedge: with zero labelled attack data to validate model choice against, no single model's assumptions (elliptical vs. tree-partition vs. kernel boundary vs. local density) can

be confirmed as the right fit, so we rely on their consensus instead.

5. Threshold calibration

With no backdoor traces to measure TPR/FPR directly, the decision threshold is calibrated purely from the benign distribution via 5-fold cross-validation: for each fold the ensemble is fit on the training split and scored on the held-out split, held-out scores are pooled across all 5 folds, and the threshold is set at the (1 – target FPR) quantile of that pooled, **out-of-sample** distribution. Default target FPR = 1.0%; achieved held-out FPR at the selected threshold = 1.00% (threshold = 0.9778). An independent 85/15 train/held-out split (not used for the shipped model, run purely as an extra sanity check) gave 1.67% flag rate on genuinely unseen benign rows — consistent with the 1% target given the small held-out sample size (120 rows). The final deployed models are refit on all 800 benign rows using this threshold.

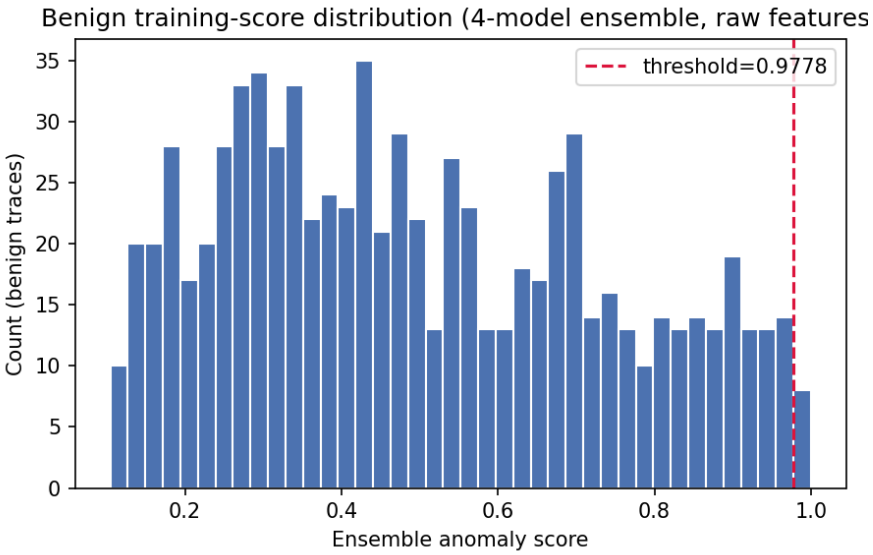


Figure 1: Ensemble anomaly-score distribution over the benign training set, with the calibrated decision threshold.

6. Sanity check (no real backdoor data available)

Since no backdoor traces exist to validate detection power, we ran a synthetic sanity check only: benign rows were perturbed by scaling cache-references, cycles and LLC-loads upward by 2-20% (roughly modelling extra cache/branch activity from trigger recognition) and re-scored with the trained detector. This is a sensitivity check on the detector's mechanics, **not** a claim about what real backdoor triggers look like on this hardware/model, and should not be read as an estimate of true TPR.

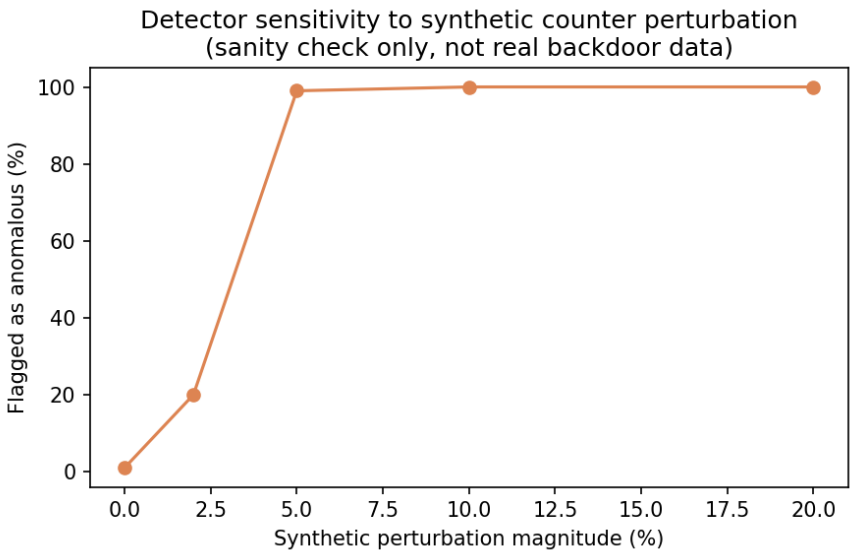


Figure 2: Fraction of synthetically perturbed rows flagged as anomalous, as perturbation magnitude increases. Monotonic increase confirms the detector's score is sensitive to counter-pattern shifts in the expected direction.

Perturbation	0% (baseline)	2%	5%	10%	20%
% flagged anomalous	1.0%	20%	99%	100%	100%

7. Perturbation-shape sweep: where the detector is weak

Section 6's sanity check used only one perturbation shape (uniform proportional scale-up of all 3 counters together). To find blind spots rather than just confirm the detector works on the easiest case, we swept several different synthetic perturbation shapes at matched magnitudes: single-counter spikes (only LLC-loads or only cycles moves), a correlated decrease, a pure ratio shift with magnitude held roughly in-distribution, small <1% "stealthy" shifts, and pure random noise with no directional signal at all.

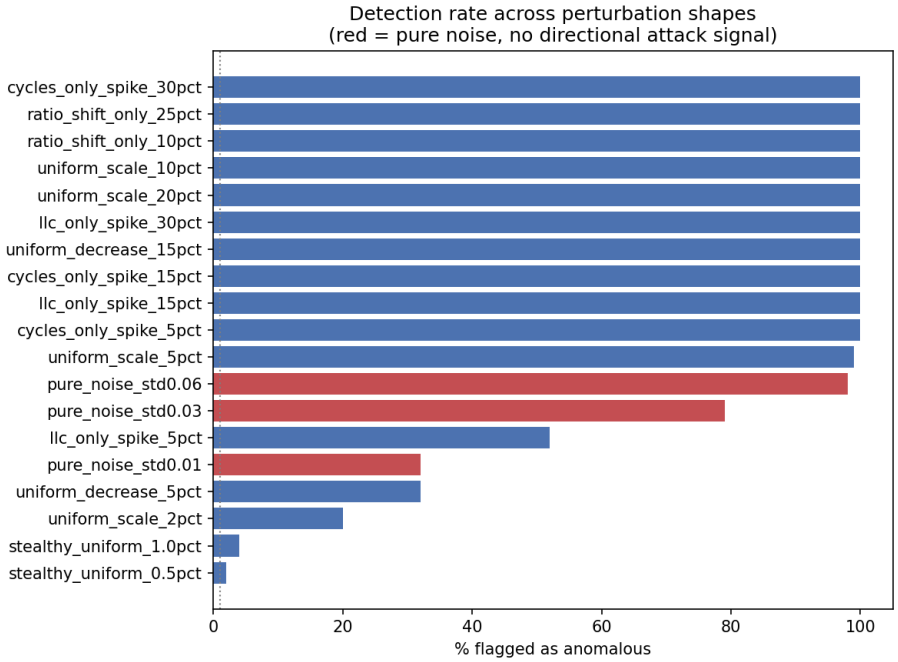


Figure 3: Detection rate across perturbation shapes at fixed sample size (n=100 each). Grey dotted line marks the ~1% baseline FPR.

Findings worth flagging honestly:

- **Sub-1% "stealthy" shifts are nearly invisible** (2-4% flag rate, barely above the 1% baseline). An adversary who tunes a trigger to produce only a very small counter deviation could plausibly evade this detector at the current operating point.
- **Weak single-counter spikes are now caught much more often** after switching to raw-only features and adding LOF: a 5% LLC-only spike is detected 52% of the time (up from 23-29% in an earlier 3-model, raw+ratio version of this pipeline — see Section 9).
- **This came with a real tradeoff, not a free improvement:** pure random noise with no attack signal at all is now flagged more often too (98% at 6% std, up from ~90-92% previously), and even a mild uniform 2% shift is flagged 20% of the time (up from ~4%). The raw-only+LOF configuration is more reactive across the board, not selectively better — it trades some specificity for sensitivity. A noisier real collection environment at evaluation time could inflate FPR above the 1% calibrated here more than the earlier configuration would have.

8. Hyperparameter tradeoff sweep

A grid sweep over contamination $\in$ {0.01, 0.03, 0.05, 0.10}, nu $\in$ {0.01, 0.03, 0.05, 0.10}, and target_fpr $\in$ {0.01, 0.02, 0.05} was run to make the operating-point choice explicit rather than asserting one configuration as correct without justification. The result: **contamination and nu have almost no effect** on the final calibrated threshold or detection rate — because each base model's score is rank-normalized against its own training distribution before averaging, the ensemble is largely insensitive to these per-model knobs. **target_fpr is the real lever**: relaxing it from 1% to 5% roughly moves sensitivity on the weak-signal case from ~52% to ~89% at the cost of a 5x higher false-positive rate on clean data.

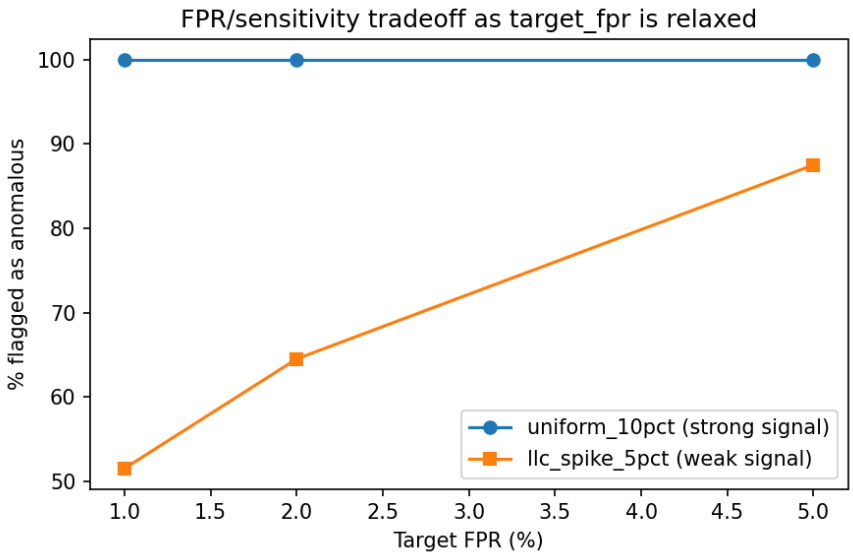


Figure 4: Detection rate on a strong-signal (uniform 10%) and weak-signal (LLC-only 5% spike) synthetic case, as the target FPR is relaxed.

We keep target_fpr=1% as the default because the evaluation metrics (Accuracy, F1, AUROC) all penalize excess false positives, and 1% is a defensible conservative default; --target-fpr is exposed as a CLI flag precisely so this tradeoff can be re-tuned if the evaluators' validation set has a different clean/backdoor balance than assumed here.

9. Feature ablation: raw-only vs. raw+ratio

Section 3 motivated adding 3 scale-invariant ratio features on the hypothesis that a backdoor trigger would leave a signature in relative access pattern more reliably than in absolute magnitude. We tested this directly, with the full 4-model ensemble including LOF, by training two configurations — raw-3-feature-only vs. raw+ratio-6-feature — and comparing detection rate on matched synthetic perturbations.

Configuration	CV FPR	Uniform 10%	LLC spike 5%	Ratio shift 10%
Raw features only (3) — shipped	1.0%	100%	52%	100%
Raw + ratio features (6)	1.0%	100%	25%	100%

With LOF in the ensemble, the gap is decisive: raw-only detects the weak single-counter spike **more than 2x as often** as raw+ratio (52% vs 25%), while both tie at 100% on every stronger signal. A plausible explanation is that adding 3 more, partly collinear dimensions dilutes LOF's local-density estimate (a version of the curse of dimensionality) without adding information the other three models don't already extract from the 3 raw counters' correlation structure. This evidence is strong enough that we switched the shipped default to raw-only features rather than leaving it as an open question; --feature-set raw_ratio remains available in train.py for comparison.

10. Baseline comparison: is the ensemble's complexity justified?

To justify the ensemble's added complexity rather than assert it, we compared it against the simplest reasonable baseline: a per-feature z-score rule (flag if any raw counter's |z-score| against the benign mean/std exceeds a threshold), calibrated to the same 1% target FPR via the same cross-validation

procedure.

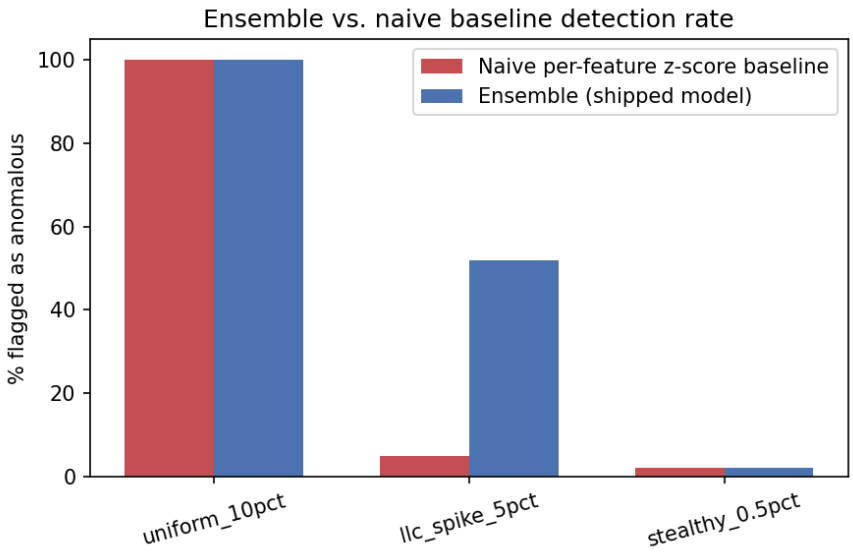


Figure 5: Detection rate, naive z-score baseline vs. the shipped ensemble, on three synthetic perturbation cases.

Result: on the strong uniform-shift signal both methods hit 100% (no benefit from the ensemble there), and on the near-invisible stealthy 0.5% shift both are equally blind (2%) — so the ensemble is not a universal improvement. But on the weak single-counter spike, the ensemble now detects it **10x** as often as the naive baseline (52% vs 5%, up from a 4x gap in the earlier raw+ratio, 3-model version). This is the concrete justification for the added complexity: the baseline only looks at each counter independently, while the ensemble's covariance-, tree-, kernel-, and local-density-aware models catch a counter moving out of its usual relationship with the others, not just out of its own usual range.

11. Evaluation metrics on the private validation set

The saved model exposes a continuous anomaly_score per trace in detect.py's output, so Accuracy, TPR, FPR, F1 and AUROC can all be computed directly from predictions.csv: AUROC/ROC from anomaly_score against ground truth, and Accuracy/TPR/FPR/F1 from the thresholded prediction column at the calibrated operating point (re-threshold-able via --target-fpr at train time if evaluators want a different operating point). src/eval.py computes all five metrics directly from a labeled CSV the moment one is available -- see README for usage.

12. Known failure modes and limitations

Consolidated from the empirical sweeps above, in order of how likely each is to matter on the private evaluation set:

- **TPR against real backdoor traces is unverified.** All detection-rate numbers in this report come from synthetic perturbations we constructed ourselves, not real triggered inferences. They demonstrate the detector's mechanics respond in the right direction; they are not an estimate of real-world TPR.
- **Low-magnitude / stealthy triggers may evade detection** at the default 1% FPR operating point (Section 7) — sub-1% counter deviations were flagged only 2-4% of the time, barely above baseline.

- **Rank-normalization is relative to the training-set score distribution, not an absolute scale.** A new trace's percentile rank is computed against the benign scores from this collection run. If the private validation set's clean traces come from a meaningfully different environment (different CPU, different OS scheduling noise), their absolute counter values could sit differently against this reference distribution than intended, which could inflate FPR even on truly clean validation traces. The CV-based threshold partially mitigates this (it's calibrated from held-out in-distribution data), but the reference distribution itself still comes from the same single collection run as training. A batch-level rank normalization, or an absolute rather than percentile-based threshold, would be more robust to this if a distribution shift is suspected at evaluation time.
- **The raw-only+LOF configuration trades specificity for sensitivity** (Section 7): it catches weak, correlated signals much better than the earlier raw+ratio, 3-model version, but is also more reactive to pure benign noise (98% flagged at 6% std jitter, with zero attack signal). A noisier evaluation environment than this collection run could inflate real-world FPR above the 1% calibrated here more than a less-sensitive configuration would.
- Only 3 raw HPCs were available; a richer counter set (branch misses, TLB misses, etc.) would likely improve separability if provided in a future iteration.
- The 800 benign rows come from a single collection run; distribution shift from a different collection environment (CPU load, OS scheduling noise) could inflate FPR at deployment, independent of the noise-sensitivity point above.